METRIC
A-A-52449
November 18, 1993
SUPERSEDING
MIL-I-62523(AT)
22 July 1987

COMMERCIAL ITEM DESCRIPTION

INSULATION, PANEL, VIBRATION-DAMPENING AND THERMAL-BARRIER

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-I-62523(AT), which is cancelled.

ABSTRACT

This CID covers requirements for insulation with vibration-dampening and thermal-barrier properties for use on panels in military vehicles.

SALIENT CHARACTERISTICS

a. Materials. The materials shall be as specified herein and in applicable drawings. The materials shall be free of defects and imperfections that might affect the serviceability and reliability of the finished product. The insulation shall have excellent resistance to petroleum, fuel, and alkalis, and shall have a specific gravity of 1.75 $\pm$ 10 percent. The insulation shall not contain any materials which would cause adverse effects on the health of the personnel. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see note d).

b. Design and construction. The design and construction of the insulation shall be such as to meet the requirements herein. Configuration, thickness, and weight of the insulation shall be as specified in the applicable drawing (see note b).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 5640

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:09Z
Check the source to verify that this is the current version before use.

c. Thermal conductivity. The thermal conductivity of the insulation shall not be greater than .3 watts per meter kelvin.

d. Indentation hardness. The indentation hardness of the insulation when using a type A durometer per ASTM D2240 shall be 85 ± 10 percent.

e. Tear resistance. The tear resistance of the insulation shall be not less than 5250 Newtons per meter.

f. Vibration-dampening properties. The vibration dampening properties of the insulation shall be determined as per ASTM E756 and shall not be less than as shown in figures 1 through 3.

g. Tensile strength. The tensile strength of the insulation shall be not less than 1.03 megapascal.

h. Flammability. The insulation shall be self-extinguishing with no burn rate.

i. Water absorption. The insulation shall have no more than a 0.1 percent weight gain when immersed in water for 24 hours, followed by 48 hours of drying.

j. Color. The color of the insulation shall be the manufacturer's dark grey, which provides the closest match to color chip number 36081 of FED-STD-595.

k. Identification markings. Identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's identification code (CAGE), the contract number, and the Engineering drawing part number (see note b).

QUALITY ASSURANCE PROVISIONS

a. Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

b. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see note b).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:09Z
Check the source to verify that this is the current version before use.

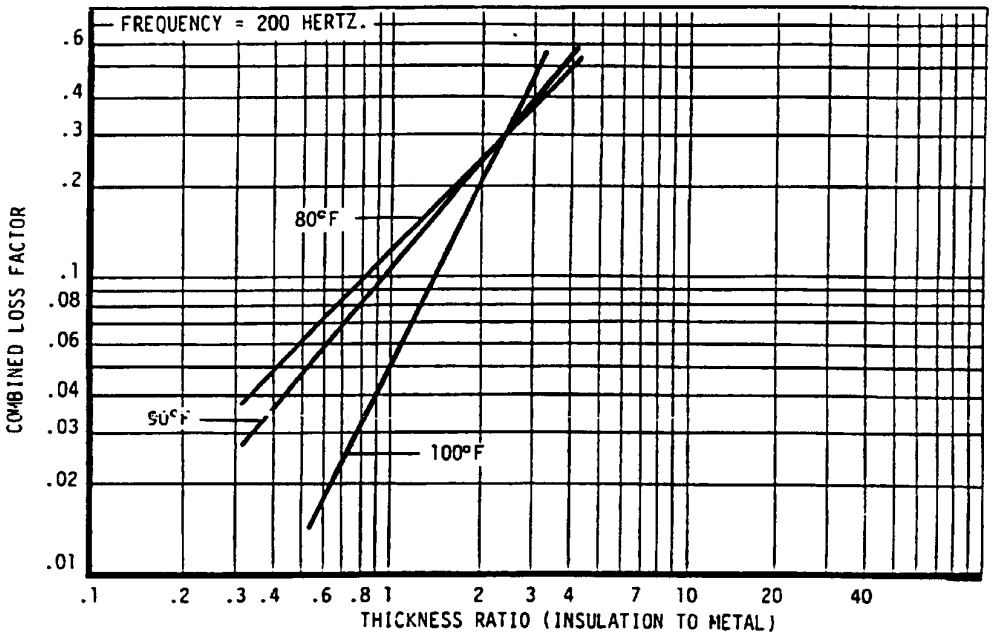


FIGURE 1. Loss factor versus thickness ratio.

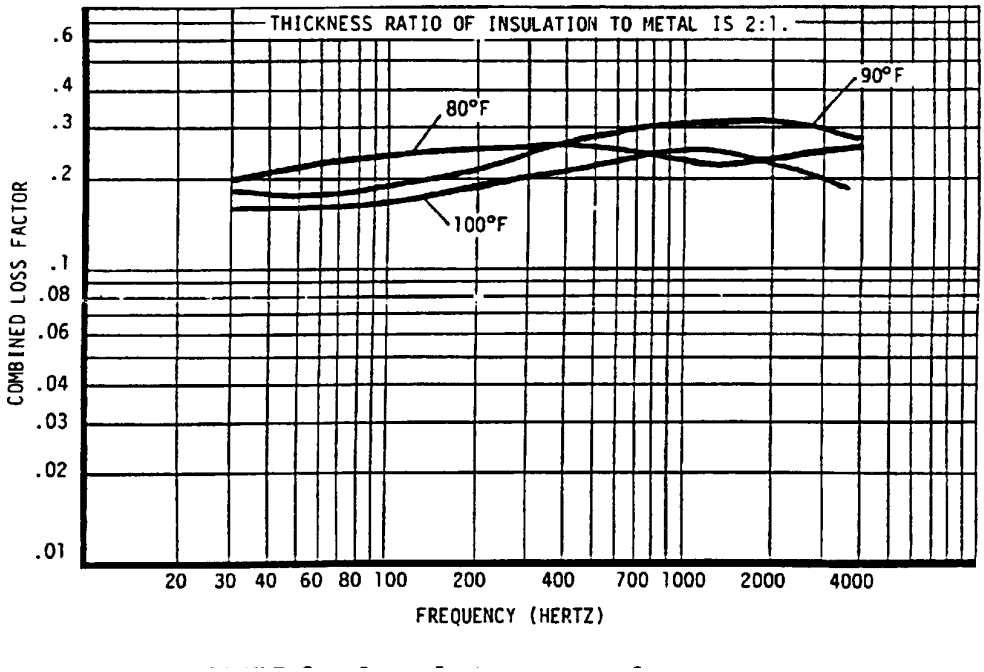


FIGURE 2. Loss factor versus frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:09Z
Check the source to verify that this is the current version before use.

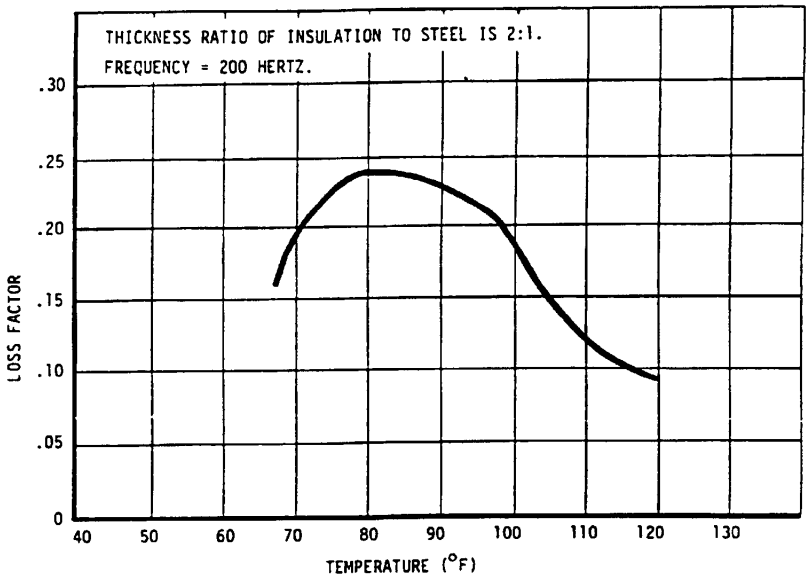


FIGURE 3. Loss factor versus temperature.

NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

a. Addresses for obtaining copies of referenced documents.

1. Government specifications and standards. Copies of FED-STD-595 “Colors Used in Government Procurement” are available from the Navy Publications and Printing Service Office, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2. Non-Government publications. Copies of ASTM E756 “Standard Test Method for Measuring Vibration-Damping Properties of Materials” and ASTM D2240 “Standard Test Method for Rubber Property-Durometer Hardness” are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.

b. Ordering data. Acquisition documents must specify the following:

1. Title, number, and date of this CID.
2. Issue of DODISS and industrial standards to be cited in the solicitation and, if required, the specific issue of individual documents referenced.
3. Title, number, and date of applicable drawing.
4. Engineering drawing part number.
5. Selection of applicable level and packaging requirements.

c. Cross-reference and supersession data. Insulation conforming to this CID are interchangeable/ substitutable with insulation conforming to MIL-I-62523(AT) which is superseded by this CID.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:09Z
Check the source to verify that this is the current version before use.

d. Regulatory requirements. Recovered material is material made in conformance to section 402 of the Clean Air Act (33 U.S.C. 1342 et. seq.).

MILITARY INTERESTS:

Custodian
Army – AT

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

(Project 5640-0563)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:09Z
Check the source to verify that this is the current version before use.